

Exploring User-Specific Variations in Vestibular Stimulation to Reduce Motion Sickness in Virtual Reality

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ABSTRACT

Motion sickness is regarded as one of important limitations in Virtual Reality (VR) applications. This study investigates the effects of vertical vibrations and additional rotational vestibular stimulation on alleviating discomfort and enhancing immersion. We conducted pilot and main studies to examine the threshold of stimulation intensity to reduce motion sickness. Experimental results revealed that the effect of vestibular stimuli varied based on user characteristics such as gender, age, and vision, highlighting the need for personalized stimulation. The initial findings offer suggestion for developing more comfortable and immersive VR experiences.

Index Terms: Virtual Reality (VR), Motion Sickness, User Immersion, Vestibular Stimulation, Motion Simulator

1 INTRODUCTION

Motion sickness is one of the main limitations in virtual reality (VR) applications, which can affect user comfort and immersion [1, 3]. Some previous studies have focused mainly on optimizing visual elements to reduce motion sickness, such as narrowing the field of view or increasing frame rates, which often compromise immersion [2]. However, this study provides a new perspective by investigating physical motion stimuli in VR. By focusing on user-specific variations, this study aims to explore personalized motion sickness mitigation, particularly in scenarios involving movements synchronized with the motion platform.

Previous studies in the field of motion platforms have focused mainly on the simulation of vehicle-based scenarios, as noted in Jung's recent publication [5]. However, our study represents an innovative contribution to the field by synchronizing motion platforms with user movements and analyzing the impact of various intensities and user-specific factors.

Unlike Groth et al. [4], who used fixed 360-degree videos with electrical stimulation to address sensory mismatches, our study integrates user movements on the motion platform (including Roll, Pitch, and Heave) with character movements to enhance immersion and comfort in VR environments. Specifically, the character leans in the direction of movement and the motion simulator adjusts accordingly to match the tilt angles (Roll and Pitch), providing varying levels of vestibular stimuli. Given that Roll and Pitch stimuli are known to induce significant motion sickness in vehicle-based environments, our study aims to evaluate their effects in a character-based VR scenario by adjusting these stimuli. Furthermore, heave-axis movements simulate up-and-down movement of the body, which is a typical movement in locomotion.

In this work, our study provides a novel approach to research on the motion platform by integrating user movements with character movements and evaluating the impact of various intensities and

user-specific factors. More specifically, we study how much vibrations on the heave axis and how varying intensities of Roll and Pitch movements along with individual user characteristics (e.g., gender, age, vision) can reduce motion sickness. We assume that personalized motion platform settings could achieve effective performance in terms of reducing motion sickness.

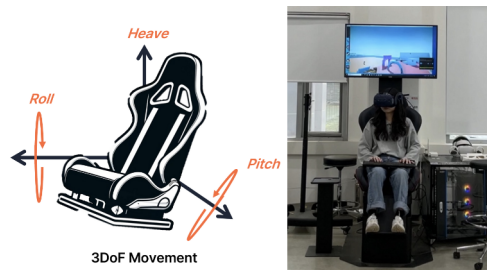


Figure 1: 3-degree-of-freedom (3DoF) system (Left) and Experimental Environment (Right)

2 METHOD

To respond to the above question, we conducted a preliminary experiment and a main experiment (Fig. 1). In both experiments, the Roll and Pitch axes were implemented as tilt movements of the motion simulator chair, providing physical tilts that align with the visual stimuli in VR. In the preliminary experiment, vibrations on the heave axis suggested the potential to reduce motion sickness, which informed the design of the main experiment. The main experiment included vibrations of the heave axis with varying tilt angles (0.5° , 2.5° , 5° , 7.5° and 10°) for the roll and pitch axes to explore the effect on motion sickness and immersion.

Thirty-two participants (16 men, 16 women) joined the main experiment, and the sample size was determined by power analysis to ensure statistically meaningful results. Each simulation session lasted approximately 5 minutes, followed by a 15-minute break based on the preliminary results of the experiment to allow participants to recover and minimize the effects of motion sickness.



Figure 2: The tasks in the virtual scene, including walking, jumping and interacting with objects.

Participants were asked to perform various tasks in a sequence, including walking, jumping, and interacting with objects, using a motion simulator synchronized with the movements of the virtual

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character (Fig. 2). The combination of motion platform movements (Roll, Pitch, and Heave) and synchronized visual cues aimed to reduce the sensory mismatch between what participants see and feel. The Heave-axis vibration was fixed in terms of both magnitude and frequency during walking to simulate the subtle up-and-down motion, while during activities like jumping or object interaction, the Heave-axis vibration varied depending on the experimental intensity settings for these tasks.

3 DATA ANALYSIS AND RESULTS

Statistical analyses included linear regression analyses to explore the relationship between motion platform intensity (Roll, Pitch, and Heave) and user response. Residuals were tested for normality using the Shapiro-Wilk test, with most following a normal distribution ($p > 0.05$). The Simulator Sickness Questionnaire (SSQ) and the Presence Questionnaire (PQ) were used to assess motion sickness and immersion, respectively [2, 3]. The MSSQ was used to establish user-specific factors based on the motion sickness history of the participants. Each user-specific factor was analyzed separately, and the results are reported individually.

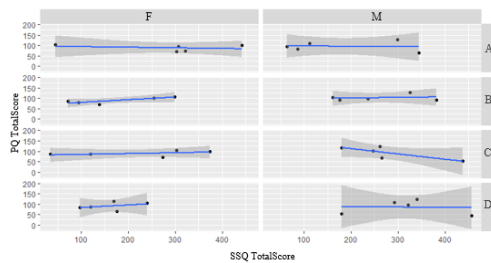


Figure 3: Differences in Motion Sickness and Immersion Scores by Gender and Tilt Intensity (A: No Movement, B: 0.5°, C: 5°, D: 10°)

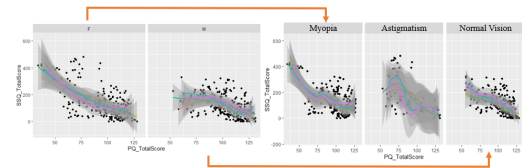


Figure 4: Comparison of Patterns between Gender and Vision Characteristics

The results indicated that there is a saturation of the disease with respect to the intensity of stimulation. However, there was no significant difference in motion sickness at all roll angles. When analyzed by user characteristics in the preliminary experiment, significant differences were observed, suggesting that individual factors may influence motion sickness (Fig. 3). Consistent trends with stimulation intensity in the main experiment suggest interactions among these characteristics (Fig. 4 and 5). In particular, women experienced lower levels of motion sickness in high-tilt conditions compared to men ($p < 0.001$). Participants with astigmatism generally reported higher levels of motion sickness ($p < 0.01$), while those with increased vestibular sensitivity demonstrated increased variability in immersion ($p < 0.001$). These initial findings suggest that the effect of motion platform movement is not uniform and that personalization could significantly enhance VR comfort and engagement.

Interestingly, the analysis also highlighted that while motion platform movement effectively reduced motion sickness, increased intensity did not necessarily correlate with improved immersion

(Fig. 5). This variability indicates that factors including prior VR experience, vestibular sensitivity, vision, gender, and age can crucially influence user experience, underlining the need for adaptive VR systems that can adjust stimuli in real time.

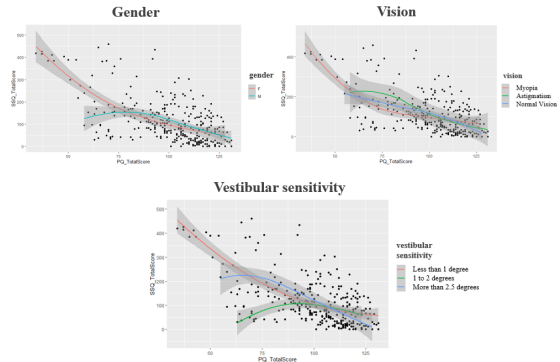


Figure 5: Similar Trends in Motion Sickness and Immersion Scores with User Characteristics (Gender, Vision, Vestibular Sensitivity)

4 CONCLUSION AND FUTURE WORK

This study provides an initial analysis of how motion platform settings (Roll, Pitch, and Heave) affect user experience to reduce motion sickness in VR. The results highlighted the importance of adapting VR design to user-specific characteristics, as individual differences could significantly impact both comfort and immersion. Future work will investigate the design of adaptive systems that adjust motion platform settings based on real-time user feedback to enhance both comfort and immersion. Furthermore, incorporating third-person perspectives into VR environments may also provide a more stable visual reference, helping to reduce motion sickness while maintaining immersion. Exploring practical applications of motion platform techniques in training, rehabilitation, and entertainment could validate these findings and enhance VR experiences.

ACKNOWLEDGMENTS

This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) under the meta-verse support program to nurture the best talents (IITP-2024-RS-2023-00256615) grant funded by the Korea government (MSIT).

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